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1. Introduction

Machine learning is a branch of artificial intelligence (AI) that concentrates on algorithms capable of “learning” the patterns from training data and thereafter, making precise predictions about new data. This skill in pattern recognition allows machine learning models to make choices or forecasts without defined, rigid commands (IBM, 2026). It encompasses supervised learning, utilizing labelled datasets to develop algorithms for precise data classification, and unsupervised learning, where it covers analysing datasets to discover hidden patterns or clusters. In this project, real world datasets will be analysed using both supervised and unsupervised learning to identify patterns and derive valuable insights.

4. Classification

4.1 Substantive issue

Breast cancer continues to be a significant worldwide health issue. In 2022, approximately 2.3 million women were diagnosed with breast cancer, resulting in 670,000 deaths globally. (WHO,2025) Correctly distinguishing breast tumours as either malignant or benign is crucial for patients and healthcare providers, as timely and precise diagnosis can enhance treatment choices and survival rates. Machine learning models can aid this process by detecting patterns in diagnostic information and enhancing prediction accuracy. Thus, using classification methods on breast cancer data could greatly enhance clinical decision-making and public health results.

4.2 Research Questions

RQ1: Which machine learning model yields the highest accuracy in categorizing breast tumours as either malignant or benign?

RQ2: What diagnostic characteristics most significantly affect the prediction of a tumour being malignant or benign?

4.3 Dataset and Variables

The breast Cancer Wisconsin (Diagnostic dataset) contains 569 entries and 30 numerical predictor variables, with no missing data found in the dataset. Every observation relates to characteristics derived from digitized images of fine needle aspirate (FNA) samples of breast masses. These attributes define the properties of the cell nuclei found in the image, including radius, texture, perimeter, area and smoothness.

The dataset includes two target categories that indicate if a tumour is malignant (212 instances) or benign (357 instances). The goal of the classification models is to precisely differentiate between these two categories using the diagnostic attributes. The dataset serves as a common benchmark in machine learning research for assessing classification algorithms. Below is a summary of chosen variables from the dataset.

Variable Name	Variable Description
Radius mean	Mean distance from the centre to points on the perimeter
Texture mean	Standard deviation of Gray-scale values
Perimeter mean	Mean size of the tumour perimeter
Area mean	Mean area of the tumour
Smoothness mean	Local variation in radius lengths
Compactness mean	Measure of tumour compactness calculated as $(\text{perimeter}^2 / \text{area} - 1.0)$
Concavity mean	Severity of concave portions of the tumour contour
Symmetry mean	Measure of symmetry of the tumour
Fractal dimension mean	Fractal dimension of the tumour boundary

Diagnosis	Tumour Classification (M = malignant, B = benign) (Target Variable)
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4.4 Methodology

An initial exploratory data analysis (EDA) will be performed on the breast cancer dataset to investigate the distribution of diagnostic variables and their associations with the tumour classification outcome. This stage aids in recognizing possible patterns, relationships among variables, and general features of the dataset before proceeding with model development. After the exploratory analysis, multiple classification algorithms will be used to determine if a tumour is malignant or benign. The training of these models will utilize the diagnostic features obtained from fine-needle aspirate images of breast tumours. The models seek to identify patterns in these features that effectively differentiate between malignant and benign tumour instances. The classification model's performance will subsequently be assessed utilizing various evaluation metrics, such as accuracy, precision, recall, F1- score, and ROC-AUC. These metrics offer an extensive evaluation of model performance by assessing both general predictive accuracy and the model's capacity to accurately pinpoint malignant tumour cases. Comparing with currently literature can offer valuable context for assessing the efficacy of machine learning models in medical diagnosis. Earlier studies have shown that diagnostic characteristics obtained from images of breast cells can be effectively utilized to distinguish tumours as malignant or benign through machine learning methods (Street, Wolberg and Mangasarian, 1993). Moreover, machine learning models like support vector machines, when integrated with feature selection, have demonstrated robust classification efficacy in breast cancer diagnostic datasets (Akay, 2009). These research efforts emphasize the capabilities of machine learning techniques in aiding breast cancer diagnosis and offer a valuable reference for assessing the models created in this research.

4.5 Analysis

Preliminary analysis was carried out with a correlation heatmap to determine the connections among the variables in the breast cancer dataset. The heatmap reveals significant positive correlations between various variables associated with tumour size. Specifically, radius mean and perimeter mean exhibit a correlation of 0.99 and area mean is also strongly correlated with perimeter mean (0.99). Robust connections are also noted among the worst-case metrics, as radius worst and perimeter worst display correlations exceeding 0.97. The strong correlations suggest that variables related to tumour size are closely linked and could offer significant predictive capability for diagnosing breast cancer, as illustrated in figure 14. The target variable's distribution was analysed to assess the dataset's class balance. The findings indicate

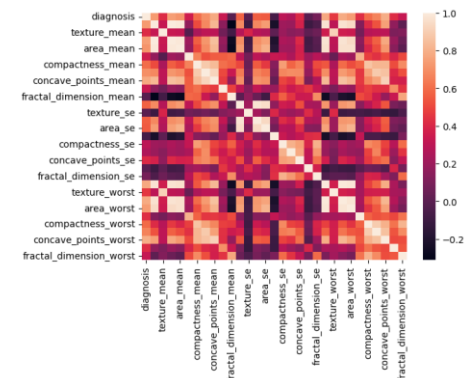


Figure 14: Correlation heatmap

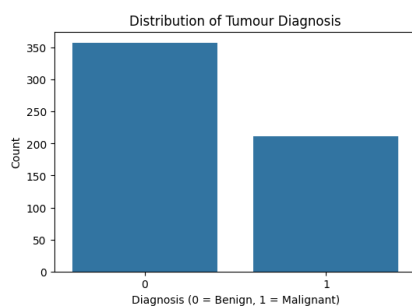


Figure 15: Distribution of Tumour

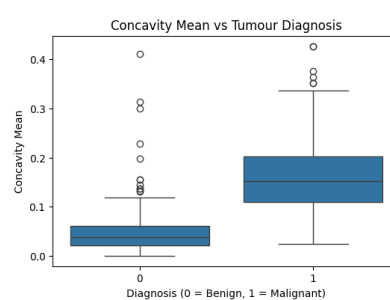


Figure 16: Radius vs Diagnosis boxplot

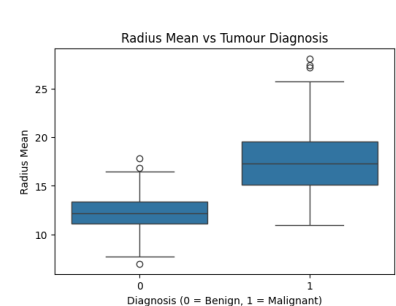


Figure 17: Area vs Diagnosis boxplot

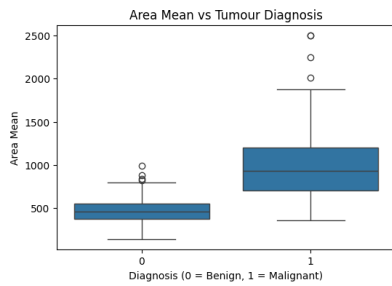


Figure 18: Concavity vs Diagnosis

carried out to explore how tumour features vary between benign and malignant instances. The boxplots indicate that malignant tumours typically exhibit larger sizes in comparison to benign tumours. For instance, malignant tumours exhibit a higher median radius mean of about 17-18, while benign tumours have a median near 12-13. A comparable trend is seen with area mean, as malignant tumours attain values greater than 1000, whereas benign tumours predominantly stay under 500. Furthermore, the concavity mean values are significantly elevated in malignant tumours, suggesting more irregular tumour edges as illustrated from figure 16 to 18. After the exploratory analysis, the dataset was readied for machine learning modelling. The dataset was divided into training and testing sets with an 80:20 distribution, yielding 455 samples for training and 114 samples for testing. Feature scaling was conducted with the StandardScaler to normalize the variables prior to model training. Subsequently, three supervised learning algorithms were employed: Logistic Regression, Random Forest and K-nearest Neighbours (KNN). Model performance was assessed using Accuracy, F1 Score, and ROC-AUC metrics, along with confusion matrices and ROC curves. The Logistic Regression model obtained the best overall performance compared to the three models. The model achieved an accuracy of 97.37% and F1 score of 0.965, and a ROC-AUC score of 0.997. The confusion matrix indicates that 70 benign cases (61.40%) and 41 malignant cases (35.96%) were accurately categorized. The testing dataset revealed just 1 false positive (0.88%) and 2 false negatives (1.75%). The Roc Curve additionally illustrates robust classification performance, with the curve situated near the upper left corner, signifying a high true positive rate as depicted in figure 19 and 20.

that the dataset contains 357 benign cases (62.7%) and 212 malignant cases (37.3%). While benign tumours are more common, the dataset keeps a balanced distribution, enabling machine learning models to efficiently discern patterns between the two classes, as illustrated in figure 15. Additional investigation was

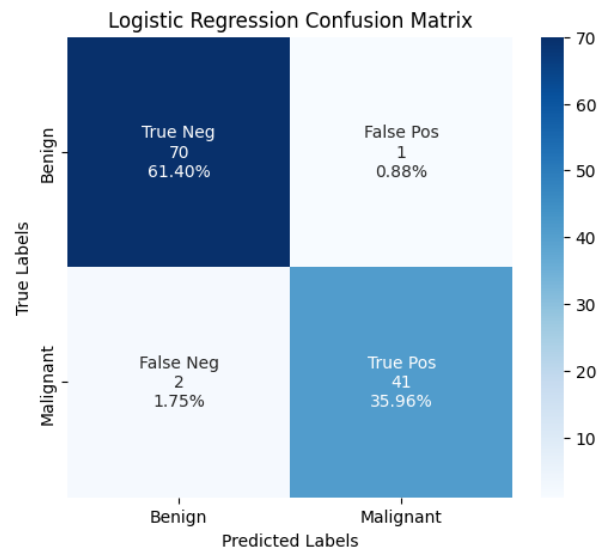


Figure 19: Logistic Regression Confusion Matrix

Logistic Regression
F1 Score: 0.965
ROC_AUC Score: 0.997
Accuracy: 0.974

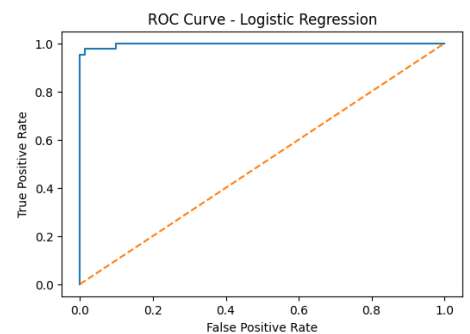


Figure 20: Logistic Regression ROC Curve

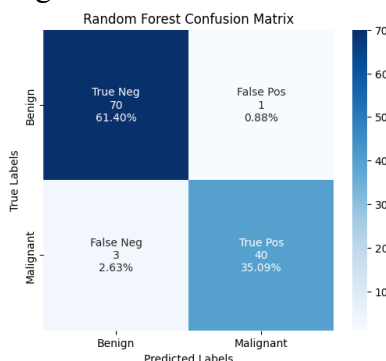


Figure 21 Random Forest Confusion

Random Forest
F1 Score: 0.952
ROC_AUC Score: 0.995
Accuracy: 0.965

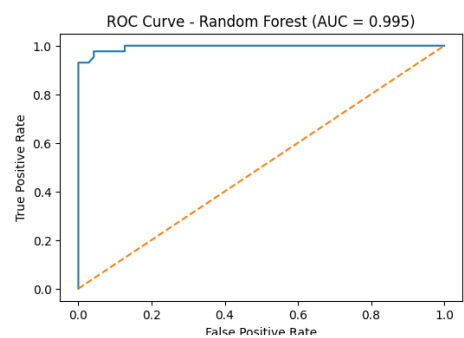


Figure 22: Random Forest ROC Curve

The Random Forest model demonstrated excellent predictive capability, achieving an accuracy of 96.49%, an F1 score of 0.952, and a ROC-AUC score of 0.995. The confusion matrix shows that 69 benign instances (60.53%) and 41 malignant instances (35.96%) were accurately identified. The analysis of feature importance indicates that area worst (0.156) and concave points worst (0.145) are the key variables in forecasting breast cancer diagnosis. Additional significant predictors are concave points mean (0.107) and radius worst (0.078), indicating that tumour size and irregular tumour edges are crucial signs of malignancy, as illustrated in figure 21.

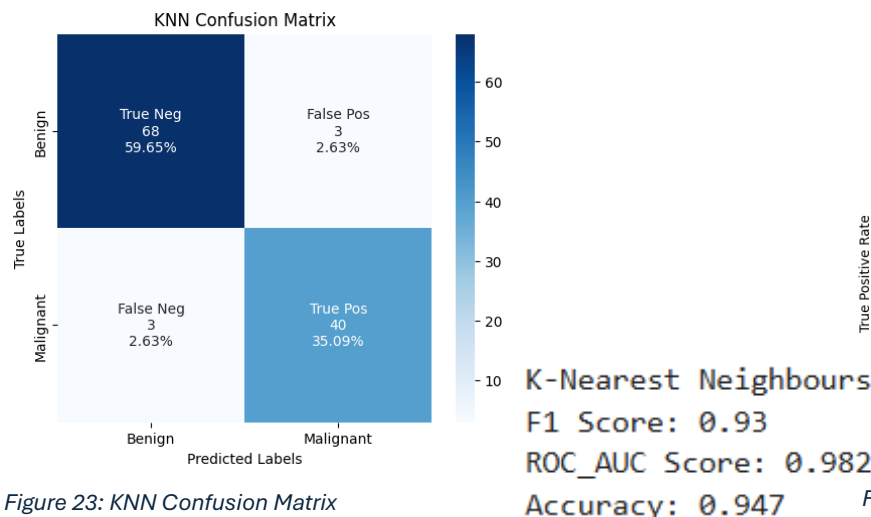


Figure 23: KNN Confusion Matrix

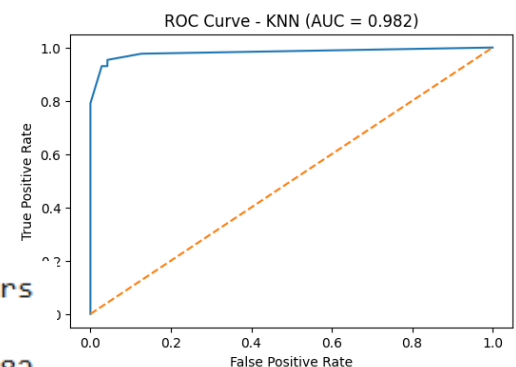


Figure 24: KNN ROC Curve

The K-Nearest Neighbour (KNN) algorithm attained an accuracy of 94.74%, along with an F1 score of 0.930 and a ROC-AUC score of 0.982. The confusion matrix indicates that 68 benign cases (59.65%) and 40 malignant cases (35.09%) were accurately categorized. Nonetheless, the model generated a few more misclassifications than the other models, recording 3 false positives (2.63%) found in the test set. Nevertheless, the ROC curve still demonstrates robust classification performance with an AUC value of 0.982.

	Model	Accuracy	F1 Score	ROC-AUC
0	Logistic Regression	0.974	0.965	0.997
1	Random Forest	0.965	0.952	0.995
2	KNN	0.947	0.930	0.982

In conclusion, a comparison among the three machine learning models indicates that logistic Regression delivered the best overall performance, with the highest accuracy (97.37%), F1 score (0.965), and ROC-AUC score (0.997). Random Forest yielded similar

results with an accuracy of 96.49% whereas KNN recorded the least performance with an accuracy of 94.74%. In this dataset, Logistic Regression demonstrated the most reliable predictive performance for breast cancer classification as shown in the model comparison table.

5. Reference

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6. Dataset Links

Classification

<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>